

US Household Income

Which states and areas have the highest and lowest household income, and how the ranking changes depending on whether you use the mean or the median.

Tools SQL. Data cleaning, joins, window functions, aggregation

Data US household income statistics. Two tables joined on area ID, 32,526 areas across 50 states, DC and Puerto Rico

The questions

- Which states have the most land and water area?
- Which states have the highest and lowest household income?
- Does the ranking change depending on whether the mean or the median is used?
- How does income differ by type of area, such as city, town, village or borough?
- Which cities have the highest average household income?

The data

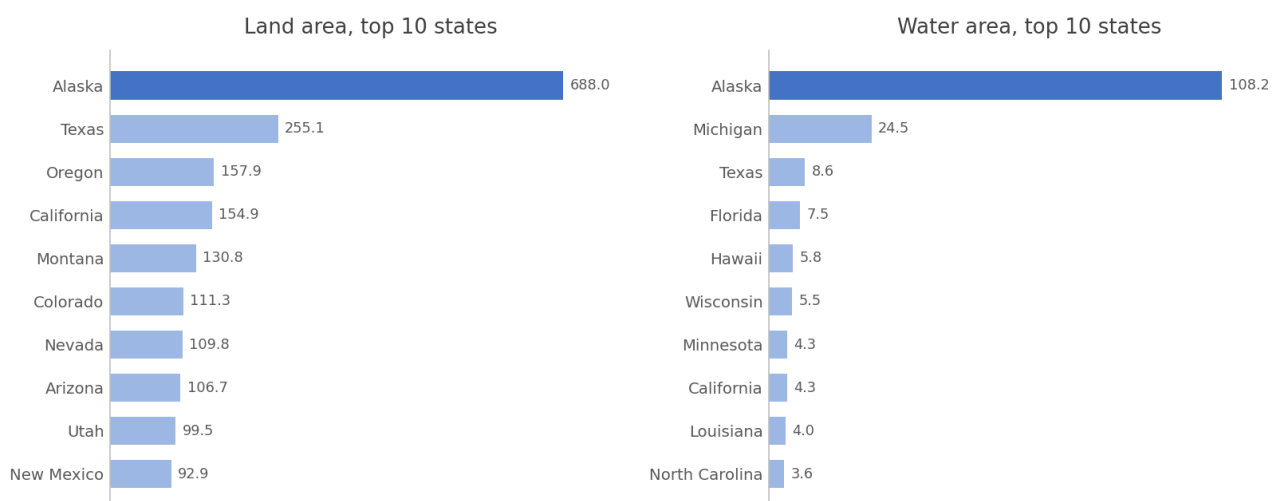
The first table describes each area: state, county, city, type of area, land and water size and coordinates. The second holds the income figures for the same areas: mean, median and standard deviation. Every ID in one table has a match in the other. Five issues across the two tables had to be resolved in SQL before the analysis.

Problem	What I did
7 duplicate area IDs	Used ROW_NUMBER() partitioned by ID to find the extra copies and delete them
State names misspelled as "georia" and "alabama"	Updated both to the correct spelling so each state groups as one
1 area with a blank place name	Filled it from the matching county and city
Type recorded both as "Borough" and "Boroughs"	Merged into a single value
315 areas with a mean income of zero	Excluded from the income analysis with a WHERE filter, leaving 32,211 areas

After deduplication both tables hold 32,526 areas. The income analysis joins them on ID.

Land and water area

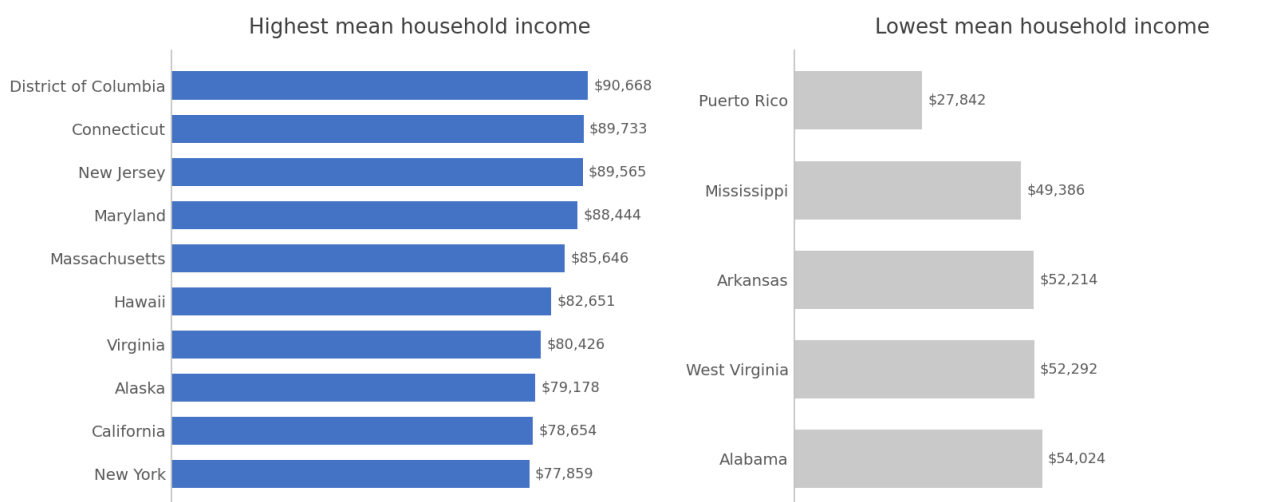
Alaska is far ahead on both measures. Its land area is about 2.7 times Texas, the next largest, and its water area is more than four times Michigan, the next largest by water. Texas, Oregon, California and Montana follow Alaska on land. Michigan, Texas, Florida and Hawaii follow on water.



Total land and water area by state, in billions of square metres. Alaska in dark blue.

Household income by state

The District of Columbia has the highest mean household income at \$90,668, followed by Connecticut, New Jersey and Maryland, all above \$88,000. Seven of the top ten are Northeast or Mid-Atlantic jurisdictions, and the other three are Hawaii, Alaska and California. Puerto Rico is at the bottom at \$27,842, well below Mississippi, the lowest of the states at \$49,386. Excluding Puerto Rico, the highest state is roughly 1.8 times the lowest.

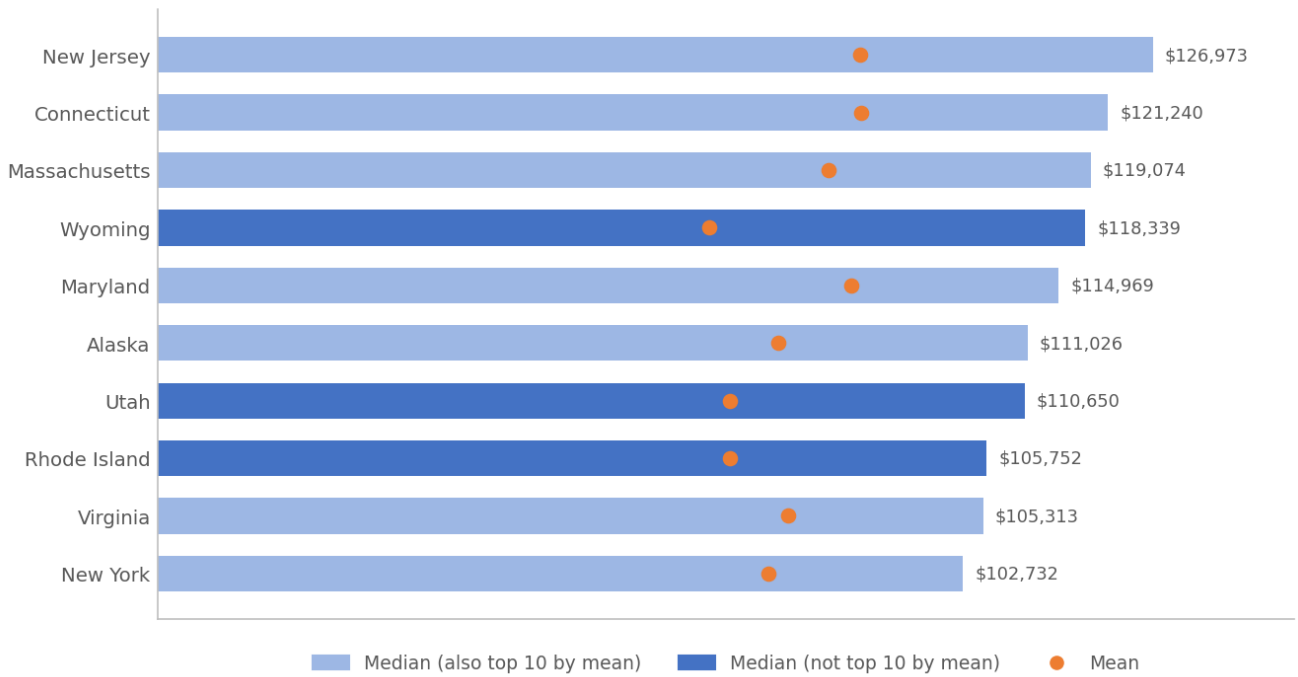


Mean household income by state. Each figure is the simple average of the area-level means within that state.

Mean versus median

Ranking by median gives a different top ten. Wyoming, Utah and Rhode Island enter the list, while the District of Columbia, Hawaii and California drop out. New Jersey moves to first place at \$126,973, with Connecticut and Massachusetts close behind. Wyoming is the clearest case: fourth by median at \$118,339 but outside the top ten by mean at \$70,367.

Top 10 states by median household income

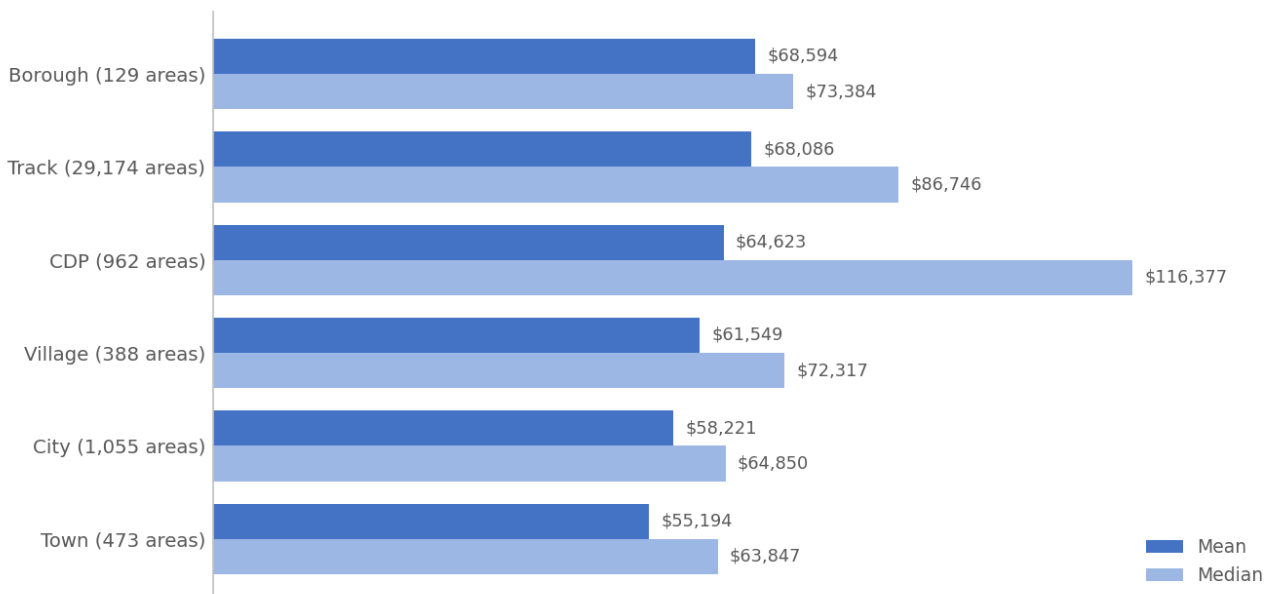


Median household income for the ten highest states, with the mean shown as a dot. Dark bars are states that are not in the top ten by mean.

Income by type of area

Borough areas have the highest mean at \$68,594, just ahead of Track areas at \$68,086. By median the order changes: CDP areas lead at \$116,377, followed by Track at \$86,746. Town areas are lowest on both measures. Smaller types were left out of the final query because they hold too few areas to compare. Community areas, for example, show the lowest mean of all at \$18,992, but there are only 17 of them.

Average household income by type of area



Mean and median household income by type of area. Only types with more than 100 areas are shown. Track areas make up 91% of the data.

Highest-income cities

The final query ranks cities by average mean income. Delta Junction in Alaska comes first at \$242,857, followed by Short Hills in New Jersey at \$216,503.

City	State	Mean household income	Areas behind the figure
Delta Junction	Alaska	\$242,857	1
Short Hills	New Jersey	\$216,503	1
Narberth	Pennsylvania	\$194,426	1
Chevy Chase	Maryland	\$194,158	2
Darien	Connecticut	\$192,882	1
Great Falls	Virginia	\$192,104	2
Pelham	New York	\$190,909	1
Chantilly	Virginia	\$190,865	1
McLean	Virginia	\$188,414	3
Bronxville	New York	\$188,005	2

Interpretation and limitations

The state and area-type rankings depend on which measure is used, so any statement about which places are "richest" should say whether it refers to the mean or the median. The two columns do not order places the same way, and for area types the gap is large.

Four limits apply. State and type figures are simple averages across areas, not weighted by population, so they describe the typical area rather than the typical household. The city ranking rests on one to three areas per city, so it is a list of high-income areas rather than a reliable ranking of cities. The last query does not apply the zero-income filter, although the top ten is the same either way. And in 51 of the 52 states the median is higher than the mean, the reverse of what income data usually shows, so the two columns should be read as separate measures from the source and not as a skew comparison.

What is in the file

Section	Contents
Data cleaning	Duplicate detection and deletion with a window function, state name and type corrections, blank place fill, zero-income and zero-area checks
Data analysis	Land and water totals by state, an inner join of the two tables, state averages of mean and median income (lowest five, top ten by mean, top ten by median), area type averages, and city averages

Portfolio project built on a public US household income dataset.